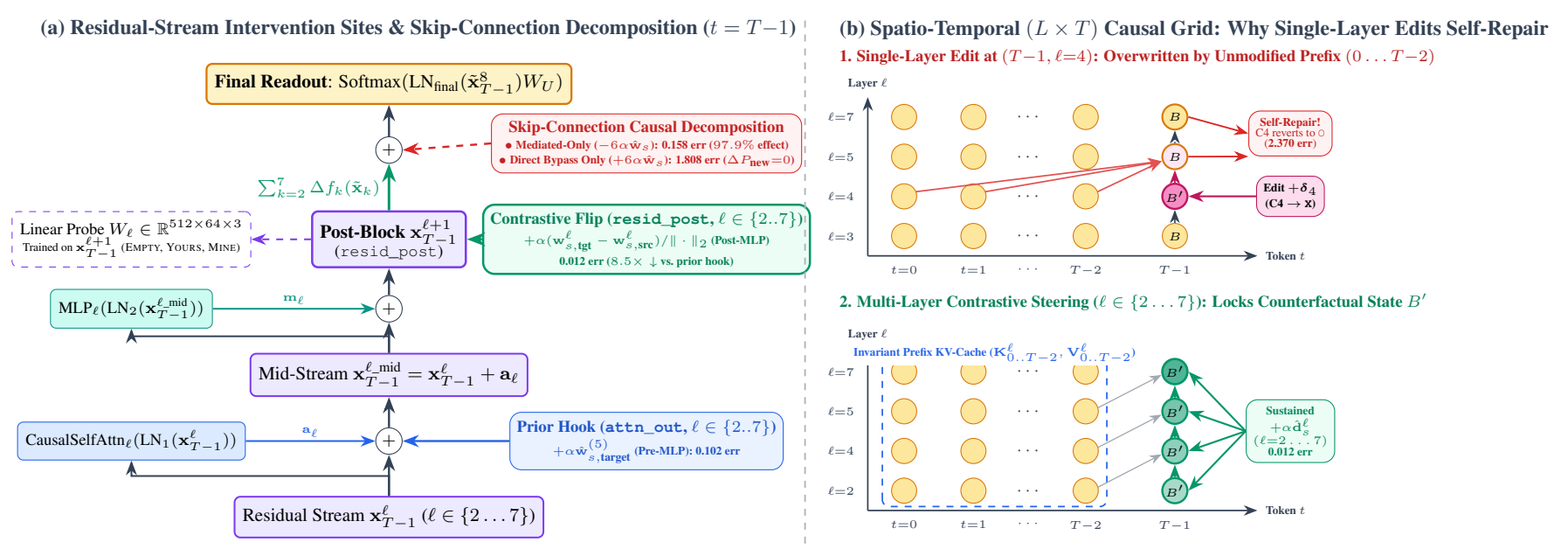

(a) Residual-Stream Intervention Sites & Skip-Connection Decomposition ($t = T-1$)
Final Readout: Softmax(LN$_{\text{final}}$($\tilde{\mathbf{x}}^{8}_{T-1}$)$W_U$)
Skip-Connection Causal Decomposition
• Mediated-Only ($-6\alpha\hat{\mathbf{w}}_s$): 0.158 err (97.9% effect)
• Direct Bypass Only ($+6\alpha\hat{\mathbf{w}}_s$): 1.808 err ($\Delta P_{\text{new}}=0$)
$\sum_{k=2}^{7}\Delta f_k(\tilde{\mathbf{x}}_k)$
Linear Probe $W_\ell \in \mathbb{R}^{512\times64\times3}$
Trained on $\mathbf{x}^{\ell+1}_{T-1}$ (EMPTY, YOURS, MINE)
Post-Block $\mathbf{x}^{\ell+1}_{T-1}$ (resid_post)
Contrastive Flip (resid_post, $\ell \in \{2..7\}$)
$+\alpha(\mathbf{w}^{\ell}_{s,\text{tgt}} - \mathbf{w}^{\ell}_{s,\text{src}})/\|\cdot\|_2$ (Post-MLP)
0.012 err (8.5× ↓ vs. prior hook)
MLP$_\ell$(LN$_2$($\mathbf{x}^{\ell_\text{mid}}_{T-1}$))
$\mathbf{m}_\ell$
Mid-Stream $\mathbf{x}^{\ell_\text{mid}}_{T-1} = \mathbf{x}^{\ell}_{T-1} + \mathbf{a}_\ell$
CausalSelfAttn$_\ell$(LN$_1$($\mathbf{x}^{\ell}_{T-1}$))
$\mathbf{a}_\ell$
Prior Hook (attn_out, $\ell \in \{2..7\}$)
$+\alpha\hat{\mathbf{w}}^{(5)}_{s,\text{target}}$ (Pre-MLP): 0.102 err
Residual Stream $\mathbf{x}^{\ell}_{T-1}$ ($\ell \in \{2\ldots7\}$)
(b) Spatio-Temporal ($L \times T$) Causal Grid: Why Single-Layer Edits Self-Repair
1. Single-Layer Edit at ($T-1$, ℓ=4): Overwritten by Unmodified Prefix ($0\ldots T-2$)
Layer ℓ
ℓ=7
ℓ=5
ℓ=4
ℓ=3
B
B
B′
B
Self-Repair!
C4 reverts to ○
(2.370 err)
Edit $+\delta_4$
(C4 → x)
t=0
t=1
$T-2$
$T-1$
Token t
2. Multi-Layer Contrastive Steering ($\ell \in \{2\ldots7\}$): Locks Counterfactual State B'
Layer ℓ
Invariant Prefix KV-Cache ($\mathbf{K}^{\ell}_{0..T-2}$, $\mathbf{V}^{\ell}_{0..T-2}$)
ℓ=7
ℓ=5
ℓ=4
ℓ=2
B′
B′
B′
B′
Sustained
$+\alpha\hat{\mathbf{a}}^{\ell}_{s}$
(ℓ=2…7)
0.012 err
t=0
t=1
$T-2$
$T-1$
Token t